

Homework 5

EE 999 — Introduction to Control Systems

Patrick Gordon • October 10, 2026

Due Wednesday, October 14, 2026

Problem 1.A — Transfer function

$$U(s) = k_p R(s) \quad (1)$$

$$Y(s) = \frac{b_1}{s + a_1} U(s) \quad (2)$$

$$Y(s) = \frac{k_p b_1}{s + a_1} R(s) \quad (3)$$

$$G_{ry1}(s) = \frac{Y(s)}{R(s)} = \frac{k_p b_1}{s + a_1} \quad (4)$$

Problem 1.B — Transfer function

$$Y(s) = \frac{b_1}{s + a_1} U(s) \quad (5)$$

$$U(s) = \frac{k_p s + k_i}{s} R(s) \quad (6)$$

$$Y(s) = \frac{b_1(k_p s + k_i)}{s^2 + a_1 s} R(s) \quad (7)$$

$$G_{ry2}(s) = \frac{Y(s)}{R(s)} = \frac{b_1(k_p s + k_i)}{s^2 + a_1 s} \quad (8)$$

Problem 1.C.1 — Transfer function

$$Y(s) = \frac{b_1}{s + a_1} U(s) \quad (9)$$

$$U(s) = \frac{k_p s + k_i}{s} E(s) \quad (10)$$

$$E(s) = R(s) - Y(s) \quad (11)$$

$$Y(s) = \frac{b_1(k_p s + k_i)}{s^2 + a_1 s} E(s) \quad (12)$$

$$Y(s) = \frac{b_1(k_p s + k_i)}{s^2 + a_1 s} (R(s) - Y(s)) \quad (13)$$

$$\left(1 + \frac{b_1(k_p s + k_i)}{s^2 + a_1 s}\right) Y(s) = \frac{b_1(k_p s + k_i)}{s^2 + a_1 s} R(s) \quad (14)$$

$$\frac{s^2 + (a_1 + b_1 k_p)s + b_1 k_i}{s^2 + a_1 s} Y(s) = \frac{b_1(k_p s + k_i)}{s^2 + a_1 s} R(s) \quad (15)$$

$$Y(s) = \frac{b_1 k_p s + b_1 k_i}{s^2 + (a_1 + b_1 k_p)s + b_1 k_i} R(s) \quad (16)$$

$$T_{ry3}(s) = \frac{Y(s)}{R(s)} = \frac{b_1 k_p s + b_1 k_i}{s^2 + (a_1 + b_1 k_p)s + b_1 k_i} \quad (17)$$